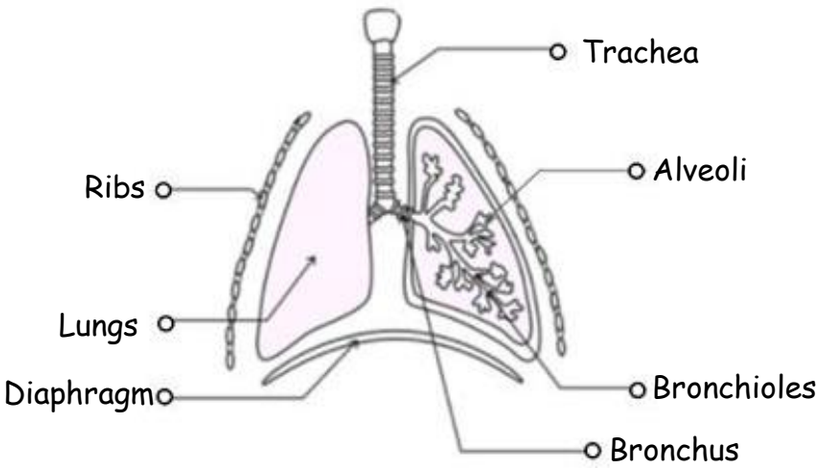
Knowledge Organiser
Year 8: Human Body – Breathing

How does breathing work?
- Breathing is a physical process that is controlled by muscular contractions
- There is a strong band of muscle in the chest called the diaphragm
- When the diaphragm contracts pressure changes in the chest pull air into the lungs

A labelled diagram of the lungs showing the breathing apparatus and gas exchange systems



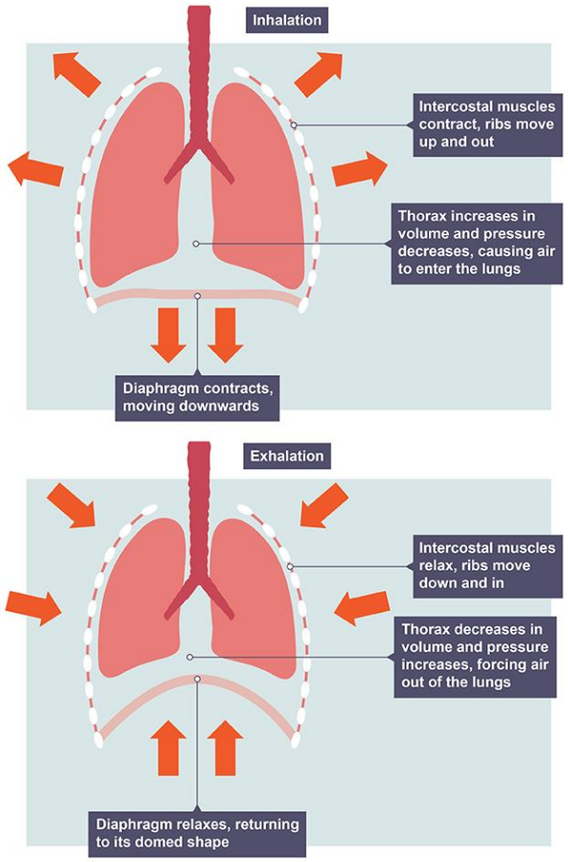
How do pressure changes cause breathing?
- When the **diaphragm contracts** (gets flatter) it pulls downwards
- This increases the volume in the chest cavity
- The increased volume leads to a decrease in **air pressure** in the lungs
- Air rushes into the lungs to counteract the change, this is called **inhaling**
- To exhale the diaphragm must **relax** and move upwards
- This increases **air pressure** in the lungs and the air is pushed back out

Did you know?

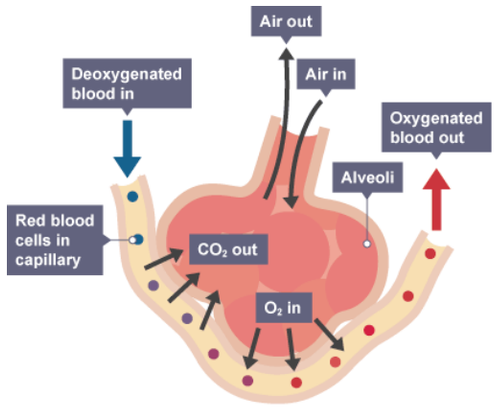
An adult has around 600 million alveoli in their lungs with a combined surface area the size of a tennis court.



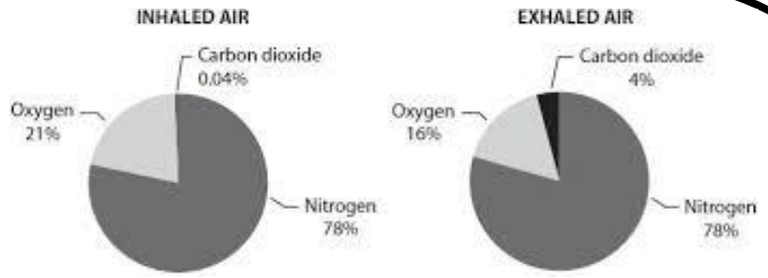
Two diagrams showing the process of breathing in terms of pressure changes in the chest



How is gas exchanged in the lungs?
- Air which is inhaled into the lungs has a **higher concentration of oxygen**
- The air is forced into the **alveoli (air sacs)** at end of the bronchioles
- The surface of the alveoli is covered in **blood vessels**
- The oxygen will then **diffuse** from the air into the blood, where the concentration is lower
- Carbon dioxide moves in the other direction, diffusing from the blood and into the alveoli
- This air is then **exhaled** and leaves the body



A labelled diagram showing how gases are exchanged in the alveoli



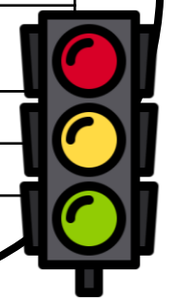
A pair of pie charts comparing the composition of inhaled and exhaled air

How is breathing different to respiration?
- Breathing is not the same as respiration
- Breathing is a **physical process** which is controlled by muscles
- Respiration is a **chemical process** which is caused by a chemical reaction in the **mitochondria**
- Breathing only happens in the **chest and lungs**
- Respiration happens in **every cell** in the body
- Breathing **brings air** into and out of the lungs
- Respiration **releases energy** from glucose

	Respiration	Breathing
What kind of process?	Chemical	Physical
Where does it happen?	In all cells	In the lungs and chest
What does it do?	Releases energy from glucose	Brings air into and out of the lungs
What controls it?	Mitochondria	Diaphragm

RAG Yourself!

Learning Outcomes;	RAG
Label the main parts of the lungs on a diagram	
Explain the adaptations show by the alveoli	
Compare breathing and respiration	
Explain how breathing works in terms of pressure in the chest	
Explain the effect of smoking on lung function	
Compare inhaled to exhaled air	

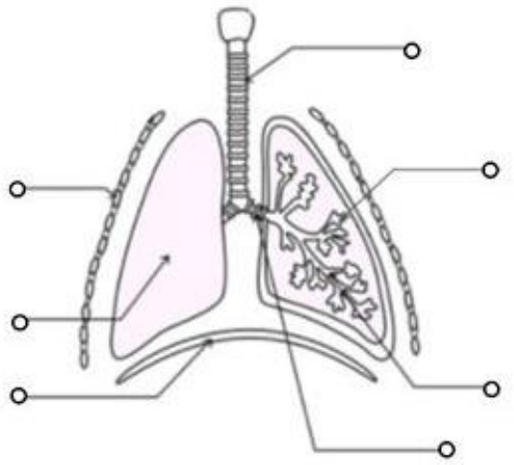


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Knowledge Organiser
Year 8: Human Body – Breathing

How does breathing work?
- B

A labelled diagram of the lungs showing the breathing apparatus and gas exchange systems



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Did you know?

An adult has around 600 million alveoli in their lungs with a combined surface area the size of a tennis court.



Two diagrams showing the process of breathing in terms of pressure changes in the chest

How is gas exchanged in the lungs?

A labelled diagram showing how gases are exchanged in the alveoli

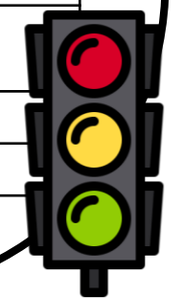
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